

Algorithm

- (1) Start the program.
- (2) Get the values of resources and processes.
- (3) Get the avail value.
- (4) After allocation find the need value.
- (5) Check whether its possible to allocate.
- (6) If it is possible then the system is in safe state.
- (7) Else system is not in safety state.
- (8) If the new request comes then check that the system is in safety.
- (9) Or not if we allow the request.
- (10) Stop the program.
- (11) End

(VII) Recourses Required

- Computer System with basic configuration. Linux/Ubuntu/ CentOS /any other open source Operating System.
- Software: Turbo C/ vi editor/ Any open source Python IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(VIII) Precautions to be followed

- (1) Handle computer system with care.
- (2) Strictly follow the instructions for writing, compiling, and executing the program.
- (3) Don't forget to save file before execution.
- (4) Follow safety Practices

(IX) Program Code

Write a C/Python program to implement Banker's Algorithm for deadlock avoidance.

Note: Attach the code at the end.

(This practical can be performed in any of the compiler like Turbo C/ vi editor etc or Python Interpreter / IDE like any open source IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(X) Conclusion

Successfully implemented C/Python program to implement Banker's algorithm

(XI) Practical Related Questions

Note : Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

- (1) What is meant by deadlock
- (2) What is safe state in banker's algorithms?
- (3) What is banker's algorithm?
- (4) What happens if a resource request leads to an unsafe state?
- (5) What are the necessary conditions where deadlock occurs?

[Space to Write for Answers]

1) A deadlock is a situation in an operating system where two or more processes are permanently blocked because each process is waiting for a resource that is being held by another process. As a result, none of the processes can continue executing.

- 2) A safe state is a state in which the system can allocate resources to all processes in some order (called a safe sequence) so that every process can complete its execution without causing a deadlock.
- 3) Banker's algorithm is a deadlock avoidance algorithm used by the operating system to allocate resources safely. Before granting a resource request, it checks whether the system will remain in a safe state. If granting the request would lead to an unsafe state, the request is postponed.
- 4) If a resource request leads to an unsafe state, the Banker's Algorithm denies or postpones the request. The process is kept waiting until the resources can be allocated while keeping the system in a safe state, thereby avoiding deadlock.
- 5) A deadlock can occur only if all four of the following conditions are present :-
 1. Mutual execution
At least one resource is non-shareable, meaning only one process can use it at a time.
 2. Hold and wait
A process holds one or more resources while waiting for additional resources held by other processes.
 3. No Preemption
Resources cannot be forcibly taken from a process; they must be released voluntarily after the process completes.

4. Circular wait

A circular chain of processes exists where each process is waiting for a resource held by the next process in the chain.